Ethan Savar

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Education

New York University — New York, NY
M.S. in Computer Science — *May 2028*

The Ohio State University — Columbus, OH
B.S. in Computer Science Engineering, B.S. in Mathematics — *May 2026*

Experience

Amazon — Sep. 2026 - Nov. 2026
Incoming Software Development Engineer Intern — *Seattle, WA*

Capital One — June 2026 - Aug. 2026
Software Engineer Intern — *McLean, VA*

- Engineered Python and Spark pipelines to detect free trial and subscription acitvity from card transactions
- Built event-driven AWS Lambda and DynamoDB workflows to notify cardholders of detected charges
- Provisioned Lambda and data infrastructure via CloudFormation and automated CI/CD with Jenkins pipelines

JPMorganChase — June 2025 - Aug. 2025
Software Engineer Intern — *Columbus, OH*

- Built a full-stack analysis tool for identifying recurring server-side failure signals for trading workflow applications, gathering data from multiple sources into a single dashboard, enabling a reduction in response to incidents by 18%
- Developed real-time APIs using Java and Spring Boot, and React frontend displaying graphs, filtering, and search
- Created a configurable health-check and SQL probe runner, enabling teams to create/parameterize queries and validate service redundancy leading to a reduction in manual verification by 3-4 hours/week

Immuta — May 2024 - May 2025
Research Scientist/Engineer Intern — *Columbus, OH*

- Developed a regex to automata compiler in Python using NFA/DFA subset conversion and Hopcroft minimization, reducing sensitive data discovery detection error rate below 1% and improving coverage by 36% across data classes
- Reduced AI Copilot costs by 12% by introducing RAG context caching and prompt compression in AWS Bedrock
- Created hierarchical semantic buckets for regex using BERT embeddings and agglomerative clustering in PyTorch
- Built a LLM prompt evaluation tool to quantify output drift across identical prompts through embedding cosine similarity and KS tests

Projects

RAG Supply Chain Assistant | *Python, Azure, Ignition, FastAPI* — Aug. 2025 – Apr. 2026

- Built and deployed a production Azure RAG assistant, implementing document chunking, embedding, and vector retrieval to ground LLM responses in supply chain data, saving hours of operational downtime per week
- Developed a FastAPI retrieval pipeline for embedding ingestion, relevance ranking, and inference, supporting real-time operator workflows in a production environment

AutoTex: Language to LaTeX | *React, Tailwind, Electron, Ollama, Python, PyTorch* — Mar. 2025 – Feb. 2026

- Built a cross-platform desktop app that converts natural language to structured LaTeX using LLM-based parsing
- Integrated a lightweight CNN/OCR model in PyTorch to extract and interpret handwritten notes and PDFs, achieving ∼94% character-level accuracy on custom datasets

Employment Outreach Locator | *Python, Tableau* — Nov. 2025

- Collaborated and won(∼20 teams) at JPMC Data For Good Hackathon by designing a Tableau dashboard and Python ranking algorithm to identify U.S. regions for nonprofit intervention targeting disadvantaged students
- Developed a composite metric leveraging job market density, GDP, university distribution, and population demographics, enabling data-driven prioritization of outreach areas

Technical Skills

Languages: Python, Java, C, x86 Assembly, Typescript, SQL, HTML/CSS
Technologies: React, Spring Boot, AWS, Spark, PyTorch, Tableau, Azure, Git, Ignition, Flask, JUnit, Jenkins